

Ring Systems — Technical Specifications

Required new systems and modifications for magic ring implementation.

Project: `/home/ubuntu/dnd35prototype`

System 1: Ring Equipment Framework (FOUNDATION — Required First)

Priority: CRITICAL

Effort: 2-3 days

Files Modified: ItemData.cs, Inventory.cs, ItemDatabase.cs, RecalculateStats()

Files Created: RingDatabase.cs, RingFactory.cs, RingDefinition.cs

1.1 ItemData Additions

```
// New fields on ItemData.cs
public bool IsRing;           // True if this item is a ring
public string RingId;         // Unique ring ID (e.g.,
"ring_of_protection_1")
public int RingBonusValue;    // Numeric bonus (+1 to +5 for protection,
etc.)
public string RingEnergyType; // For energy resistance rings:
"acid"/"cold"/"electricity"/"fire"/"sonic"
public RingEffectType RingEffect; // What type of ring this is
public bool RingIsActive;      // For toggle rings (Force Shield)
public string RingStoredSpellId; // For Ring of Counterspells
public int RingStoredSpellLevel; // For Ring of Counterspells
public int RingWizardryLevel;  // 1-4 for Ring of Wizardry
```

1.2 RingEffectType Enum

```
public enum RingEffectType
{
    None,
    Protection,           // Deflection bonus to AC
    EnergyResistance,     // Resist energy damage
    Evasion,              // Grant Evasion ability
    FreedomOfMovement,    // Continuous FoM
    ForceShield,          // Toggle +2 shield AC
    MindShielding,        // Immunity to mental detection
    Invisibility,         // Command word → Invisibility spell
    Blinking,             // Command word → Blink spell
    Telekinesis,          // Command word → Telekinesis spell
    Regeneration,         // Fast healing 1/round
    Ram,                  // Charged ranged force attack
    AnimalFriendship,     // Command word → Charm Animal
    ChameleonPower,       // +10 Hide + Disguise Self
    FeatherFalling,       // Auto Feather Fall
    Sustenance,           // No food/water needed
    SkillBonus,           // +5/+10 to specific skill
    WaterWalking,         // Walk on water
    XRayVision,           // See through walls
    MeldIntoStone,       // Command word → Meld into Stone
    Wizardry,            // Double arcane spell slots
    Counterspells,       // Store + auto-counter
    SpellStoring,         // Multi-spell storage
    SpellTurning,         // Daily spell reflection
    ShootingStars,        // Multiple daily abilities
    FriendShield,         // Paired Shield Other
    ThreeWishes,          // 3x Wish charges
    DjinniCalling,        // Summon djinni
    ElementalCommand,     // Elemental dominance + spells
}
```

1.3 RingDefinition Class

```
public class RingDefinition
{
    public string RingId;           // Unique identifier
    public string Name;             // Display name
    public string Description;      // Tooltip text
    public int MarketPrice;        // GP value
    public int CasterLevel;        // CL of ring
    public string AuraSchool;      // Aura school
    public string AuraStrength;    // Faint/Moderate/Strong
    public RingEffectType EffectType; // What the ring does
    public int BonusValue;         // Numeric bonus (if applicable)
    public string EnergyType;      // Energy type (if applicable)
    public int MaxCharges;         // For charged rings (0 = unlimited)
    public bool IsExpendable;      // Becomes nonmagical when charges = 0
    public string SpellId;         // For spell-like rings
    public int SpellCasterLevel;   // CL for spell effect
    public string ActivationType;  // "continuous", "command", "use-activated"
    public RingImplementationStatus Status; // Implementation tracking
}

public enum RingImplementationStatus
{
    NotStarted,
    Stub,           // Registered but effect not implemented
    Partial,        // Some effects work
    Complete        // Fully functional
}
```

1.4 Inventory.RecalculateStats() Ring Processing

Add after existing armor/weapon processing:

```
// — Ring Effects —
ProcessRingEffects(LeftRingSlot);
ProcessRingEffects(RightRingSlot);

private void ProcessRingEffects(ItemData ring)
{
    if (ring == null || !ring.IsRing) return;

    switch (ring.RingEffect)
    {
        case RingEffectType.Protection:
            // Take highest deflection bonus (ring vs spell)
            OwnerStats.DeflectionBonus = Mathf.Max(
                OwnerStats.DeflectionBonus, ring.RingBonusValue);
            break;

        case RingEffectType.EnergyResistance:
            // Add energy resistance entry
            AddRingEnergyResistance(ring);
            break;

        case RingEffectType.Evasion:
            OwnerStats.HasEvasion = true;
            break;

        case RingEffectType.ForceShield:
            if (ring.RingIsActive)
                OwnerStats.ShieldBonus = Mathf.Max(OwnerStats.ShieldBonus, 2);
            break;

        // ... other passive effects
    }
}
```

1.5 ItemDatabase.CloneItem() Update

Add ring field copying (same pattern as staff fields):

```
clone.IsRing = src.IsRing;
clone.RingId = src.RingId;
clone.RingBonusValue = src.RingBonusValue;
clone.RingEnergyType = src.RingEnergyType;
clone.RingEffect = src.RingEffect;
clone.RingIsActive = src.RingIsActive;
clone.RingStoredSpellId = src.RingStoredSpellId;
clone.RingStoredSpellLevel = src.RingStoredSpellLevel;
clone.RingWizardryLevel = src.RingWizardryLevel;
```

System 2: Command Word Activation (Required for Tier 2)

Priority: HIGH

Effort: 2 days

Files Modified: GameManager.cs (TryUseItem flow), CombatUI.cs

Files Created: RingActivationPanel.cs (or extend QuickItemUsePanel)

Design

When a player clicks “Use Item” and selects an equipped ring with command word activation:

1. GameManager detects `currentItem.IsRing && ring.ActivationType == "command"`
2. Shows activation UI (like staff spell selection but simpler)
3. Consumes standard action
4. Applies ring spell effect using existing spell resolution

Integration Points

- `GameManager.TryUseItem()` — add `else if (currentItem.IsRing)` branch
- Reuse existing spell effect application from wand/staff system
- Ring of Invisibility → cast Invisibility at ring’s CL
- Ring of Blinking → cast Blink at ring’s CL
- Ring of Telekinesis → cast Telekinesis at ring’s CL

System 3: Ring Charge Tracking (Required for Charged Rings)

Priority: MEDIUM

Effort: 1 day (reuse wand pattern)

Files Modified: ItemData.cs (reuse CurrentCharges/MaxCharges from wand system)

Design

Ring of Ram and Ring of Three Wishes use charges. This already exists for wands:

```
// Already on ItemData:
public int CurrentCharges;
public int MaxCharges;
```

For rings, add:

- Ring of Ram: MaxCharges = 50, charge cost selection UI (1/2/3)
- Ring of Three Wishes: MaxCharges = 3, each use = 1 charge
- When charges hit 0 and `IsExpendable == true`, convert to nonmagical ring

System 4: Spell Slot Doubling (Required for Ring of Wizardry)

Priority: HIGH (high gameplay impact)

Effort: 3 days

Files Modified: SpellSlotCalculator.cs (or wherever spell slots are computed), CharacterStats.cs

Design

Ring of Wizardry doubles BASE arcane spell slots for a specific level:

```
// In spell slot calculation:
int baseSlots = GetBaseArcaneSpellSlots(spellLevel);
int abilityBonusSlots = GetAbilityBonusSlots(spellLevel);

// Check for Ring of Wizardry
int wizardryLevel = GetEquippedRingOfWizardryLevel(); // 0 if none
if (wizardryLevel == spellLevel && IsArcaneClass)
{
    baseSlots *= 2; // Double base slots only
}

int totalSlots = baseSlots + abilityBonusSlots;
```

Key Rules

- ONLY arcane casters (Wizard, Sorcerer, Bard)
- ONLY base spell slots doubled (not ability bonus slots)
- Cannot double slots for levels the caster can't cast yet
- If two Rings of Wizardry for same level: doesn't triple (same bonus type)

System 5: Spell Storage (Required for Ring of Spell Storing + Counterspells)

Priority: MEDIUM

Effort: 4-5 days

Files Created: SpellStorageData.cs, SpellStorageUI.cs

Files Modified: ItemData.cs, GameManager.cs

Design

```
public class StoredSpell
{
    public string SpellId;
    public int SpellLevel;
    public int CasterLevel;
    public int SaveDC;
    public string CasterName;
}

// On ItemData:
public List<StoredSpell> StoredSpells; // For Ring of Spell Storing
public int SpellStorageCapacity;      // 3/5/10 total spell levels
```

Storage Flow

1. Caster casts a spell "into" the ring (uses spell slot, no target)
2. Spell stored with original caster's CL, DC, etc.
3. Total spell levels cannot exceed capacity

Casting Flow

1. Ring wearer activates ring (standard action)
2. UI shows stored spells

3. Select spell → cast at original CL/DC
4. Spell removed from storage

Counterspell Flow (Ring of Counterspells)

1. Only stores ONE spell (level 1-6)
2. When that exact spell targets wearer → auto-counter
3. No action required from wearer
4. Stored spell consumed
5. Can reload

System 6: Daily Use Tracking (Required for Shooting Stars, Spell Turning)

Priority: LOW-MEDIUM

Effort: 2 days

Files Modified: ItemData.cs, GameManager round tracking

Design

```
// On ItemData:
public Dictionary<string, int> DailyUsesRemaining; // ability name → uses left today
public Dictionary<string, int> WeeklyUsesRemaining; // ability name → uses left this week

// Reset on long rest:
public void ResetDailyUses() { ... }
public void ResetWeeklyUses() { ... }
```

Ring of Shooting Stars Daily Abilities

```
"dancing_lights": 1/hour (track separately)
"light": 2/night
"ball_lightning": 1/night
"shooting_stars": 3/week
"faerie_fire": 2/day
"spark_shower": 1/day
```

System 7: Elemental Dominance (Required for Ring of Elemental Command)

Priority: LOW (Tier 4)

Effort: 5-7 days

Files Created: ElementalDominanceSystem.cs, ElementalCommandRing.cs

Design

Complex system with:

1. **Activation tracking** — ring starts as lesser form, requires killing elemental

2. **Elemental type detection** — identify creatures by elemental subtype
3. **Passive bonuses** — attack/save bonuses vs elementals
4. **Spell-like abilities** — 6 per ring variant, varied daily limits
5. **Weakness** — save penalty vs opposing element

Data Structure

```
public class ElementalCommandRingState
{
    public bool IsFullyActivated;
    public string ElementType; // "air", "earth", "fire", "water"
    public Dictionary<string, int> AbilityUsesRemaining;
    // Tracks: gust_of_wind: 2/day, wind_wall: unlimited, etc.
}
```

Summary: New Systems by Priority

#	System	Effort	Tier Unlocks	Priority
1	Ring Equipment Framework	2–3 days	ALL rings	CRITICAL
2	Command Word Activation	2 days	Tier 2 (9 rings)	HIGH
3	Ring Charge Tracking	1 day	Ram, Three Wishes	HIGH
4	Spell Slot Doubling	3 days	Wizardry I–IV	HIGH
5	Spell Storage	4–5 days	Spell Storing, Counterspells	MEDIUM
6	Daily Use Tracking	2 days	Shooting Stars, Spell Turning	MEDIUM
7	Elemental Dominance	5–7 days	Elemental Command x4	LOW

Total new system effort: ~19–23 days

File Impact Map

New Files (8)

Assets/Scripts/Equipment/RingDefinition.cs	☐ Ring data structure
Assets/Scripts/Equipment/RingDatabase.cs	☐ Ring registry (all 33 types)
Assets/Scripts/Equipment/RingFactory.cs	☐ Ring item generation
Assets/Scripts/Equipment/RingEffectType.cs	☐ Ring effect enum
Assets/Scripts/Equipment/SpellStorageData.cs	☐ Spell storage for rings
Assets/Scripts/UI/Panels/RingActivationPanel.cs	☐ Ring use UI
Assets/Scripts/UI/Panels/SpellStorageUI.cs	☐ Spell store/retrieve UI
Assets/Scripts/Systems/ElementalDominanceSystem.cs	☐ Elemental command framework

Modified Files (10+)

Assets/Scripts/Inventory/ItemData.cs	☐ Ring fields
Assets/Scripts/Inventory/Inventory.cs	☐ RecalculateStats() ring processing
Assets/Scripts/Inventory/ItemDatabase.cs	☐ CloneItem() ring fields
Assets/Scripts/Core/GameManager.cs	☐ TryUseItem() ring branch
Assets/Scripts/Core/SceneBootstrap.cs	☐ RingDatabase.Init()
Assets/Scripts/Character/CharacterStats.cs	☐ Ring-derived stats
Assets/Scripts/Identifiers/ItemIDs.cs	☐ Ring item ID constants
Assets/Scripts/UI/PreCombatInventoryUI.cs	☐ Ring tooltip updates
Assets/Scripts/UI/InventoryUI.cs	☐ Ring display
Assets/Scripts/Magic/SpellSlotCalculator.cs	☐ Wizardry doubling

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